ANDROID PROGRAMMING PRACTICAL 10

Module 2 · SQLite database CRUD operations

Aim

Implement SQLite database CRUD operations.

Project name: SQLiteCRUDDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a form with fields and buttons for Insert, View, Update, and Delete.

2. Create a DBHelper extending SQLiteOpenHelper and define a table in onCreate().

3. Implement insert using ContentValues and writable database.

4. Implement read using a Cursor and close the Cursor after iteration.

5. Implement update and delete using selection arguments rather than concatenating untrusted input into SQL.

6. Connect the buttons to DBHelper methods and refresh the displayed records after each change.

7. Run the app and test create, read, update, and delete cases including missing IDs or empty values.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

Records can be inserted, read, updated, and deleted from SQLite.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.